

The Width-Four q -Fibonomial Coefficients Are Unimodal

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Abstract

Bergeron, Ceballos, and Küstner conjectured that every q -Fibonomial coefficient is unimodal. Recent work proves the cases of width at most three. We prove the next infinite family: for every integer $m \geq 1$, the polynomial $\left[\begin{smallmatrix} m+4 \\ 4 \end{smallmatrix} \right]_{\mathcal{F}}$ is unimodal. After taking a first coefficient difference, the problem reduces below the midpoint to four translates of the restricted partition function for parts 1, 2, 3. Its period-six quasipolynomial formula yields elementary quadratic lower bounds for all $m \geq 8$; the seven remaining cases are checked exactly.

1 Introduction

Let $F_0 = 0$, $F_1 = 1$, and $F_{j+2} = F_{j+1} + F_j$. For a positive integer r , write

$$[r]_q = 1 + q + \cdots + q^{r-1}, \quad [F_r]_q! = \prod_{j=1}^r [F_j]_q.$$

Building on the unweighted path-domino model of Sagan and Savage [3], Bergeron, Ceballos, and Küstner define

$$\left[\begin{smallmatrix} m+n \\ n \end{smallmatrix} \right]_{\mathcal{F}} = \frac{[F_{m+n}]_q!}{[F_m]_q! [F_n]_q!}. \quad (1)$$

Their path-domino model proves that (1) is a polynomial with nonnegative integer coefficients [1, Theorem 2.4]; they conjecture that all these polynomials are unimodal [1, Conjecture 2.5]. Connelly, Ito, Martinez, Shevchenko, and Yang prove the conjecture for $n \leq 3$ [2, Theorem 1.2] and report finite computations for larger parameters.

This note proves the width-four case. The proof is algebraic and uses the published path-domino theorem only to import polynomiality and nonnegativity. The May 2026 preprint [2] explicitly leaves the cases $n \geq 4$ open and notes that its proposed product criterion does not directly fit their factorization. Our literature search through August 6, 2026 found no later proof of width four; this is a bounded search statement, not an exhaustive priority claim.

Theorem 1. *For every integer $m \geq 1$, the q -Fibonomial coefficient $\left[\begin{smallmatrix} m+4 \\ 4 \end{smallmatrix} \right]_{\mathcal{F}}$ is unimodal.*

Here a polynomial $C(q) = \sum_{j=0}^D c_j q^j$ is *unimodal* if its coefficient sequence first weakly increases and then weakly decreases.

2 The midpoint difference formula

Fix $m \geq 1$, and put

$$a = F_{m+1}, \quad b = F_{m+2}.$$

The next two Fibonacci numbers are $a + b$ and $a + 2b$. Since $[F_1]_q = [F_2]_q = 1$, $[F_3]_q = [2]_q$, and $[F_4]_q = [3]_q$, equation (1) becomes

$$W_m(q) := \left[\begin{matrix} m+4 \\ 4 \end{matrix} \right]_{\mathcal{F}} = \frac{[a]_q [b]_q [a+b]_q [a+2b]_q}{[2]_q [3]_q}. \quad (2)$$

Its degree is

$$D = 3a + 4b - 7. \quad (3)$$

Indeed, $[r]_{q^{-1}} = q^{1-r}[r]_q$, so (2) gives $W_m(q^{-1}) = q^{-D}W_m(q)$. The path-domino theorem makes W_m a polynomial, and its constant term is one; consequently it has degree D and is symmetric.

For an integer t , define

$$p(t) = [q^t] \frac{1}{(1-q)(1-q^2)(1-q^3)}.$$

Lemma 2. *One has*

$$p(t) = \begin{cases} \left\lfloor \frac{t^2 + 6t + 12}{12} \right\rfloor, & t \geq 0, \\ 0, & t < 0. \end{cases} \quad (4)$$

In particular, for $t \geq 0$,

$$\frac{t^2 + 6t + 1}{12} \leq p(t) \leq \frac{t^2 + 6t + 12}{12}. \quad (5)$$

Proof. For $t \geq 0$, the generating function shows that $p(t)$ counts the solutions of $x + 2y + 3z = t$ in nonnegative integers. Thus

$$p(t) = \sum_{z=0}^{\lfloor t/3 \rfloor} \left(\left\lfloor \frac{t-3z}{2} \right\rfloor + 1 \right).$$

Writing $t = 6u + r$, $0 \leq r < 6$, and summing separately over even and odd z gives (4) in each of the six residues. The upper bound in (5) is immediate. Since the numerator in (4) is an integer, rounding down loses at most $11/12$, which gives the lower bound. $\square$

Write $W_m(q) = \sum_{t=0}^D c_t q^t$, with $c_{-1} = 0$, and let $T = \lfloor D/2 \rfloor$. Symmetry reduces unimodality to showing $c_t - c_{t-1} \geq 0$ for $0 \leq t \leq T$. From (2),

$$(1-q)W_m(q) = \frac{(1-q^a)(1-q^b)(1-q^{a+b})(1-q^{a+2b})}{(1-q)(1-q^2)(1-q^3)}. \quad (6)$$

Lemma 3. *For $0 \leq t \leq T$,*

$$c_t - c_{t-1} = p(t) - p(t-a) - p(t-b) + p(t-(2a+b)). \quad (7)$$

Proof. Expand the numerator of (6) by inclusion-exclusion. The pair shift $a+b$ cancels the singleton shift $a+b$, while the pair shift $b+(a+b) = a+2b$ cancels the singleton shift $a+2b$. The only other pair shift that can occur at or below T is $a+(a+b) = 2a+b$. Indeed, every remaining pair shift is at least $2a+2b > T$, as follows directly from (3). Every triple shift is also at least $2a+2b > T$. The four surviving terms give (7). $\square$

3 Uniform positivity

We now prove the required coefficient inequalities for $m \geq 8$. In this range, $a \geq F_9 = 34$. Let the right side of (7) be $g(t)$.

If $0 \leq t < a$, then $g(t) = p(t) > 0$. If $a \leq t < b$, then $g(t) = p(t) - p(t-a) \geq 0$, since the counting function p is nondecreasing.

Suppose next that $b \leq t < 2a+b$. The last term of (7) vanishes. Applying the lower estimate in (5) to the positive term and the upper estimate to the two negative terms gives

$$12g(t) \geq K(t), \quad (8)$$

where

$$K(t) = -t^2 + (2a + 2b - 6)t - a^2 - b^2 + 6a + 6b - 23.$$

The quadratic K is concave, so its minimum on the integer interval $[b, 2a+b-1]$ occurs at an endpoint. We have

$$K(b) = a(2b - a + 6) - 23,$$

and

$$K(2a+b-1) > K(2a+b) = a(2b - a - 6) - 23, \quad (9)$$

where the first difference is exactly $2a+5$. Since $b > a$ and $a \geq 34$, the displayed expressions for $K(b)$ and $K(2a+b)$ are positive. Thus $g(t) > 0$ throughout this interval.

Finally, suppose $2a+b \leq t \leq T$. Applying the appropriate side of (5) to all four terms of (7) yields

$$12g(t) \geq -2at + 3a^2 + 4ab - 6a - 22. \quad (10)$$

The right side decreases with t . By (3),

$$t \leq T \leq \frac{3a + 4b - 7}{2},$$

and substitution in (10) gives

$$12g(t) \geq a - 22 > 0. \quad (11)$$

This proves all midpoint differences for $m \geq 8$.

4 The seven finite cases

For $1 \leq m \leq 7$, we evaluate the exact integer formula (7) over the entire interval $0 \leq t \leq T$. The resulting minima are shown in Table 1.

m	(a, b)	T	$\min_{0 \leq t \leq T} g(t)$
1	(1,2)	2	0
2	(2,3)	5	0
3	(3,5)	11	0
4	(5,8)	20	0
5	(8,13)	34	1
6	(13,21)	58	1
7	(21,34)	96	1

Table 1: Exact finite cases not covered by the uniform estimate.

Every listed minimum is nonnegative. Together with the preceding section, Lemma 3 shows that the coefficients weakly increase to the midpoint for every $m \geq 1$. Symmetry then proves Theorem 1.

5 Reproducibility and claim boundary

The finite calculation uses only integer arithmetic. From the repository root, it is replayed by

```
python3 proof/qfib_width4_unimodality_proof.py
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The script performs three checks: it compares (4) with direct enumeration for $0 \leq t \leq 300$; it verifies every entry of Table 1; and, as an independent consistency check, it constructs the exact quotient (2) for $1 \leq m \leq 24$ and compares all midpoint differences with (7). The last two ranges beyond $m = 7$ are checks, not assumptions in the proof.

The theorem proves only the family $\left[\begin{smallmatrix} m+4 \\ 4 \end{smallmatrix} \right]_{\mathcal{F}}$, $m \geq 1$. It does not prove the full two-parameter conjecture, a combinatorial chain decomposition, log-concavity, or any width $n \geq 5$.

AI assistance. The author used AI assistance in preparing this draft and its verification code.

References

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